

Polymer Property Prediction using Machine Learning

Detailed Project Report

1. Abstract

This project predicts two important polymer properties—Glass Transition Temperature (Tg) and Chain Band Gap (Egc)—from polymer SMILES strings. The work demonstrates how machine learning can accelerate materials research by reducing the need for expensive laboratory experiments.

2. Introduction

Polymers are used in healthcare, electronics, packaging, aerospace, automotive, and energy storage. Traditionally, evaluating polymer properties requires synthesis and laboratory testing. Machine learning enables researchers to estimate these properties before experimentation, reducing cost and development time.

3. Problem Statement

The goal is to build a regression model that predicts Tg and Egc using only the SMILES representation of polymers. The competition evaluates predictions using the average R^2 score across both targets.

4. Objectives

- Predict Tg accurately.
- Predict Egc accurately.
- Build an end-to-end ML pipeline.
- Generate a valid Kaggle submission file.

5. Dataset Description

Training data contains 6,171 rows with columns: smiles, target, and target_type (Tg/Egc). Test data contains id, smiles, and target_type. The output is a submission.csv file containing id and predicted target values.

6. Methodology

SMILES strings are converted into numerical molecular features using RDKit. Feature engineering extracts descriptors such as molecular weight, hydrogen bond counts, ring counts, and other structural properties. These features are used to train regression models.

7. Technology Stack

Language: Python

Libraries: Pandas, NumPy, RDKit, Scikit-learn, Matplotlib

Environment: Kaggle Notebook / Jupyter Notebook

8. Machine Learning Pipeline

1. Load datasets.
2. Preprocess data.
3. Generate RDKit descriptors.
4. Train regression model.
5. Validate using R^2 .
6. Predict test data.
7. Create submission.csv.

9. Evaluation Metric

The competition uses the mean coefficient of determination (R^2) across Tg and Eg. Higher values indicate better predictive performance.

10. Applications

Applications include sustainable polymer discovery, biomedical materials, flexible electronics, batteries, aerospace composites, automotive materials, and eco-friendly packaging.

11. Advantages

- Reduces laboratory cost.
- Accelerates material discovery.
- Supports sustainable research.
- Enables rapid virtual screening of polymers.

12. Limitations

Prediction quality depends on training data quality, molecular feature representation, and the chosen machine learning model. Experimental validation is still required before deployment.

13. Future Scope

Future improvements include LightGBM, CatBoost, XGBoost, Graph Neural Networks (GNNs), ChemBERTa, ensemble learning, and hyperparameter optimization.

14. Repository Structure

README.md
requirements.txt
notebook.ipynb
.gitignore
LICENSE
images/

15. Conclusion

This project demonstrates how machine learning and cheminformatics can assist researchers in predicting polymer properties from chemical structures. By combining molecular feature extraction with regression models, it provides a practical workflow for accelerating materials science research.

16. References

1. RDKit Documentation
2. Scikit-learn Documentation
3. Kaggle AISEHack 2.0 Competition Description

